US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12386173
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Nakadoi; Takahide et al.

Vibration device and imaging device

Abstract

A vibration device includes a tubular vibration body, a retainer fixed to the vibration body, and a light transmissive body to be vibrated by the vibration body and held between the vibration body and the retainer. The light transmissive body includes a first surface and a second surface opposed to the first surface. The vibration body includes one end supporting the first surface of the light transmissive body. The retainer includes a side wall including a first end and a second end surrounding an outer circumference of the light transmissive body, and a supporting portion extending inwardly with respect to the side wall from the first end of the side wall. The retainer is fixed to the vibration body at a portion in contact with the vibration body on a side of the second end. The supporting portion includes a supporting surface on a side of the second surface of the light transmissive body to support the second surface of the light transmissive body.

Inventors: **Nakadoi; Takahide (Nagaokakyo, JP), Kitamori; Nobumasa (Nagaokakyo, JP)**

Applicant: **Murata Manufacturing Co., Ltd.** (Nagaokakyo, JP)

Family ID: **81382463**

Assignee: **MURATA MANUFACTURING CO., LTD. (Kyoto, JP)**

Appl. No.: **18/135338**

Filed: **April 17, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230258925 A1	Aug. 17, 2023

Foreign Application Priority Data

JP	2020-180765	Oct. 28, 2020

Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/037390 20211008 PENDING child-doc US 18135338

Publication Classification

Int. Cl.: **G02B27/00** (20060101); **B06B1/06** (20060101); **G03B15/00** (20210101); **G03B17/02** (20210101); **G03B17/56** (20210101)

U.S. Cl.:

CPC **G02B27/0006** (20130101); **B06B1/06** (20130101); **B06B1/0655** (20130101); **G03B15/00** (20130101); **G03B17/02** (20130101); **G03B17/56** (20130101); B06B2201/77 (20130101)

Field of Classification Search

CPC: G02B (27/0006); B06B (1/06); B06B (1/0655); B06B (2201/77); G03B (15/00); G03B (17/02); G03B (17/56)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2003/0214599	12/2002	Ito	348/335	H04N 23/52
2009/0262232	12/2008	Kim	348/340	G02B 7/102
2018/0095272	12/2017	Fujimoto	N/A	G03B 17/56
2018/0117642	12/2017	Magee	N/A	G01H 1/00
2019/0033685	12/2018	Fujimoto	N/A	H04N 23/55
2020/0284741	12/2019	Magee	N/A	B06B 1/0292

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
09223910	12/1996	JP	N/A
2017170303	12/2016	JP	N/A
2019536018	12/2018	JP	N/A

OTHER PUBLICATIONS

English machine translation of Masuyama et al. JP H09223910 (Year: 1997). cited by examiner
International Search Report in PCT/JP2021/037390, mailed Dec. 21, 2021, 3 pages. cited by applicant
Written Opinion in PCT/JP2021/037390, mailed Dec. 21, 2021, 4 pages. cited by applicant

Primary Examiner: Bologna; Dominic J

Attorney, Agent or Firm: Keating & Bennett, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-180765 filed on Oct. 28, 2020 and is a Continuation Application of PCT Application No. PCT/JP2021/037390 filed on Oct. 8, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The present invention relates to a vibration device and an imaging device including the vibration device.
2. Description of the Related Art
(2) A droplet removal device which removes a droplet or the like adhering to a luminous flux passing range of a drip-proof cover drip-proof is known.
(3) For example, Japanese Unexamined Patent Application Publication No. 2017-170303 discloses a droplet removal device provided with an excitation member which is connected to an end portion of a curved surface forming a dome part of an optical element, and causes flexural vibration to the dome part.
(4) The droplet removal device described in Japanese Unexamined Patent Application Publication No. 2017-170303 controls the excitation member to remove a droplet or the like adhering to the dome part by causing vibration to be applied to the dome part and miniaturizing the droplet or the like.
SUMMARY OF THE INVENTION
(5) The droplet removal device described in Japanese Unexamined Patent Application Publication No. 2017-170303 still has room for improvement in terms of improving reliability of a joint portion.
(6) Therefore, preferred embodiments of the present invention provide vibration devices and imaging devices in which reliability of a joint portion between a light transmissive body and a vibration body is improved.
(7) A vibration device according to an aspect of a preferred embodiment of the present invention includes a tubular vibration body, a retainer fixed to the vibration body, and a light transmissive body to be vibrated by the vibration body and held between the vibration body and the retainer.
(8) An imaging device according to an aspect of a preferred embodiment of the present invention includes the vibration device described above and an imaging element inside the vibration device.
(9) According to preferred embodiments of the present invention, the vibration devices and the imaging devices in each of which reliability of the joint portion between the light transmissive body and the vibration body is improved can be provided.
(10) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a vibration device according to Preferred Embodiment 1 of the present invention.
(2) FIG. **2**A is a sectional view of the vibration device in FIG. **1**.
(3) FIG. **2**B is an enlarged view of a region R**1** in FIG. **2**A.
(4) FIG. **3** is an exploded perspective view illustrating components of the vibration device in FIG. **1**.
(5) FIG. **4** is a sectional view illustrating an imaging device including the vibration device in FIG.

1.
(6) FIG. **5** is a graph illustrating a time period taken before separation occurs at a joint portion between a light transmissive body and a vibration body in an environment at a temperature of about 85° C. and a humidity of about 85%.
(7) FIG. **6** is a sectional view of a vibration device according to Modification 1 of Preferred Embodiment 1 of the present invention.
(8) FIG. **7** is a partial sectional view illustrating a vibration device according to Preferred Embodiment 2 of the present invention.
(9) FIG. **8** is a graph illustrating relationship between a position where a supporting portion of a retainer supports the light transmissive body, and compressive stress.
(10) FIG. **9** is a schematic sectional view illustrating an inner diameter of the supporting portion of the retainer, an inner diameter of the vibration body, and an outer diameter of the light transmissive body in the vibration device in FIG. **7**.
(11) FIG. **10** is a sectional view of a vibration device according to Preferred Embodiment 3 of the present invention.
(12) FIG. **11** is a partial sectional view of a vibration device according to Preferred Embodiment 4 of the present invention.
(13) FIG. **12** is a perspective view illustrating a vibration device according to Preferred Embodiment 5 of the present invention.
(14) FIG. **13** is a partial sectional view illustrating the vibration device in FIG. **12**.
(15) FIG. **14** is an exploded perspective view illustrating the vibration device in FIG. **12**.
(16) FIG. **15** is an exploded perspective view illustrating a vibration device according to Preferred Embodiment 6 of the present invention.
(17) FIG. **16** is a partial sectional view of the vibration device in FIG. **15**.
(18) FIG. **17** is an exploded perspective view illustrating a vibration device according to Preferred Embodiment 7 of the present invention.
(19) FIG. **18** is a partial sectional view of the vibration device in FIG. **17**.
(20) FIG. **19** is a partial sectional view illustrating a vibration device according to Preferred Embodiment 8 of the present invention.
(21) FIG. **20**A is a diagram illustrating distribution of compressive stress of a vibration device of Example 2.
(22) FIG. **20**B is a diagram illustrating distribution of compressive stress of a vibration device of Example 3.
(23) FIG. **20**C is a diagram illustrating distribution of compressive stress of a vibration device of Reference Example 2.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Background of Preferred Embodiments of the Present Invention

(24) Cameras which are used outdoors (for example, car-mounted cameras, surveillance cameras, and cameras mounted on drones) are provided with a cover made of glass, transparent plastic, or the like so as to cover a lens because the camera may be weathered. When foreign matters, such as mud or oil, adhere to the cover, the foreign matters unexpectedly appear in an image captured by the camera, and a clear image is not always obtained since a field of vision of the camera is obstructed.
(25) Therefore, a device like the droplet removal device described in Japanese Unexamined Patent Application Publication No. 2017-170303 has been examined. The device removes a droplet or the like adhering to a drip-proof cover by a piezoelectric material being provided to the drip-proof cover and generating flexural vibration in the drip-proof cover and miniaturize the droplet or the like.
(26) In the droplet removal device described in Japanese Unexamined Patent Application Publication No. 2017-170303, a flange portion of the drip-proof cover and the piezoelectric

material are fixed to each other by adhesive, and a drip-proof seal formed in a circular annular shape is disposed to be closely fitted to the entire circumference of an outer circumferential edge portion of the flange portion.
(27) However, in the configuration described in Japanese Unexamined Patent Application Publication No. 2017-170303, since the drip-proof seal is made of rubber material, it cannot firmly hold the drip-proof cover and the piezoelectric material, and the drip-proof cover and the piezoelectric material may separate from each other. Particularly, separation of the drip-proof cover and the piezoelectric material is likely to occur in an environment at a high temperature and a high humidity. Therefore, improving reliability of the joint portion between the drip-proof cover and the piezoelectric material in the environment at a high temperature and a high humidity is an issue. The joint portion between the drip-proof cover and the piezoelectric material indicates a portion where the drip-proof cover and the piezoelectric material adhere to each other. Further, the reliability of the joint portion means unlikeliness of separation and high strength at the joint portion between the drip-proof cover and the piezoelectric material.
(28) Further, there is a problem that, since the drip-proof seal has a property of absorbing vibration, it becomes a cause of vibration loss upon transmitting vibration of the piezoelectric material to the drip-proof cover.
(29) Therefore, the present inventors examined a configuration which improves reliability of a joint portion between a cover glass and a vibration body by the cover glass being held by a retainer and the vibration body, and thus conceived of and developed the following preferred embodiments of the present invention.
(30) A vibration device according to an aspect of a preferred embodiment of the present invention includes a tubular vibration body, a retainer fixed to the vibration body, and a light transmissive body to be vibrated by the vibration body and held between the vibration body and the retainer.
(31) In this configuration, the vibration device in which reliability of the joint portion between the light transmissive body and the vibration body is improved can be provided.
(32) The light transmissive body may include a first surface and a second surface opposed to the first surface, the vibration body may include one end and support the first surface of the light transmissive body at the one end, the retainer may include a side wall including a first end and a second end and surrounding an outer circumference of the light transmissive body, and a supporting portion extending inwardly with respect to the side wall from the first end of the side wall, the retainer may be fixed to the vibration body at a portion in contact with the vibration body on a side of the second end, and the supporting portion may include a supporting surface located on a side of the second surface of the light transmissive body to support the second surface of the light transmissive body.
(33) In this configuration, the vibration device in which reliability of the joint portion between the light transmissive body and the vibration body is improved can be provided. By the light transmissive body being supported while being sandwiched between the vibration body and the supporting portion of the retainer, separation at the joint portion between the light transmissive body and the vibration body can be prevented.
(34) At least one of adhesion between the one end of the vibration body and the first surface of the light transmissive body or adhesion between the supporting surface of the retainer and the second surface of the light transmissive body may be provided by adhesive.
(35) In this configuration, the joint portion between the light transmissive body and the vibration body can firmly be fixed, and separation can be prevented.
(36) The supporting portion of the retainer may include a first supporting portion extending inwardly with respect to the side wall from the first end of the retainer, and a second supporting portion projecting toward the second surface of the light transmissive body from a tip end of the first supporting portion and including the supporting surface, and a space may be provided between the first supporting portion and the second surface of the light transmissive body.

(37) In this configuration, by the second supporting portion of the retainer being provided, the light transmissive body can be supported at the inner side portion where amplitude is large. Therefore, stress applied to the joint portion between the light transmissive body and the vibration body can be reduced or prevented. Thus, reliability of the joint portion between the light transmissive body and the vibration body can further be improved.
(38) A width of the second supporting portion of the retainer in a direction in which the supporting portion extends may be about 0.3 times or larger and about 3 times or smaller a thickness of the first supporting portion, for example.
(39) In this configuration, reliability of the joint portion between the light transmissive body and the vibration body can further be improved.
(40) The light transmissive body may have a circular or substantially circular plate shape, the vibration body and the retainer may have a cylindrical or substantially cylindrical shape, and an inner diameter of the supporting portion of the retainer may be equal to or larger than an inner diameter of the vibration body and smaller than about one-half of a sum of the inner diameter of the vibration body and an outer diameter of the light transmissive body.
(41) In this configuration, a field of view of a camera inside the vibration device can be widened while reliability of the joint portion is improved.
(42) The light transmissive body may have a circular or substantially circular plate shape, the vibration body and the retainer may have a cylindrical or substantially cylindrical shape, and an inner diameter of the supporting portion of the retainer may be smaller than an inner diameter of the vibration body and smaller than about one-half of the sum of the inner diameter of the vibration body and an outer diameter of the light transmissive body.
(43) In this configuration, since a contact area between the retainer and the light transmissive body increases, the light transmissive body can be fixed further firmly.
(44) The vibration body may have a second threaded portion on a side of the one end of the vibration body, and the retainer may have a first threaded portion on the side of the second end to be threadedly engaged with the second threaded portion.
(45) In this configuration, the retainer and the vibration body can firmly be fixed to each other.
(46) The retainer and the vibration body may be fixed to each other by a plurality of screws.
(47) In this configuration, the retainer and the vibration body can be fixed to each other easily and at low cost.
(48) The supporting portion of the retainer may include one or a plurality of plate springs.
(49) In this configuration, the manufacturing cost of the vibration device can be reduced.
(50) A groove extends continuously along an inner circumference of the retainer in a surface of the supporting portion of the retainer opposed to the second surface of the light transmissive body, and a seal is provided to the groove.
(51) In this configuration, entry of foreign matters through space between the retainer and the light transmissive body can be prevented.
(52) The light transmissive body may include a body portion, and a frame portion at an outer edge of the body portion and having a thickness smaller than a thickness of the body portion, the supporting portion of the retainer may support the frame portion, and the thickness of the frame portion may be about one-tenth or larger and about nine-tenth or smaller of the thickness of the body portion.
(53) In this configuration, the field of view of the camera inside the vibration device can be widened while reliability of the joint portion between the light transmissive body and the vibration body is improved.
(54) An imaging device according to an aspect of a preferred embodiment of the present invention includes the vibration device described above, and an imaging element inside the vibration device.
(55) Hereinafter, preferred embodiments of the present invention are described with reference to the accompanying drawings. Note that the following description is not intended to limit the present

disclosure, application, or usage thereof. Further, the drawings are schematic drawings, and each dimensional ratio or the like does not always match the actual one.

Preferred Embodiment 1

(56) Overall Configuration

(57) FIG. **1** is a perspective view of a vibration device **10** according to Preferred Embodiment 1. FIG. **2**A is a sectional view of the vibration device **10** in FIG. **1**. FIG. **2**B is an enlarged view of a region R**1** in FIG. **2**A. FIG. **3** is an exploded perspective view illustrating components of the vibration device **10** in FIG. **1**. FIG. **4** is a sectional view illustrating an imaging device **100** including the vibration device **10** in FIG. **1**.

(58) As illustrated in FIG. **1**, the vibration device **10** according to Preferred Embodiment 1 includes a light transmissive body **11**, a vibration body **12**, and a retainer **13**. In this preferred embodiment, the vibration device **10** is a device which removes a water droplet or the like adhering to the light transmissive body **11** by the vibration body **12** transmitting vibration of a piezoelectric material **14** fixed to the vibration body **12** to the light transmissive body **11** and vibrates the light transmissive body **11**. Moreover, in this preferred embodiment, the vibration device **10** is provided with a conductor **15** which applies electric potential to the piezoelectric material **14**.

(59) As illustrated in FIG. **4**, the imaging device **100** according to Preferred Embodiment 1 includes the vibration device **10** and an imaging part **30**. The imaging part **30** is disposed inside the vibration device **10**. The imaging part **30** includes, for example, an optical element, an imaging element, a sensor component, and so forth, and is provided with a case component which accommodates the incorporated component. The imaging part **30** captures an image of an imaging target object located outside of the vibration device **10** through the light transmissive body **11** of the vibration device **10**.

(60) Components of the vibration device **10** will be described below in detail.

(61) Light Transmissive Body

(62) As illustrated in FIGS. **2**A and **2**B, the light transmissive body **11** includes a first surface **11***a* and a second surface **11***b* opposed to the first surface **11***a*, and has a plate shape. As illustrated in FIG. **4**, the light transmissive body **11** is a cover which prevents a foreign matter from adhering to a lens of the imaging part **30** when the imaging part **30** is disposed inside the vibration device **10**.

(63) The first surface **11***a* of the light transmissive body **11** is supported by the vibration body **12**, and the second surface **11***b* of the light transmissive body **11** is supported by a supporting portion **16** of the retainer **13**. That is, the light transmissive body **11** is supported by being sandwiched between the vibration body **12** and the supporting portion **16** of the retainer **13**. In other words, the supporting portion **16** of the retainer **13** presses the second surface **11***b* of the light transmissive body **11** in a direction intersecting with the second surface **11***b*. In detail, the supporting portion **16** of the retainer **13** presses the second surface **11***b* of the light transmissive body **11** in a direction vertical to the second surface **11***b*.

(64) As a material of the light transmissive body **11**, for example, glass (for example, soda glass, borosilicate glass, aluminosilicate glass, or quartz glass), light transmissive plastic, a light transmissive ceramic material, or synthetic resin can be used. Strength of the light transmissive body **11** can be improved by the light transmissive body **11** being made of tempered or toughened glass whose strength is improved by, for example, chemical strengthening.

(65) In this preferred embodiment, as illustrated in FIG. **3**, the light transmissive body **11** preferably has a circular or substantially circular plate shape, for example. In detail, the light transmissive body **11** preferably has a circular or substantially circular shape when seen in a thickness direction (Y direction).

(66) The first surface and the second surface of the light transmissive body **11** may be applied with coating, such as AR coating, water repellent coating, or impact resistance coating, as necessary.

(67) Vibration Body

(68) The vibration body **12** is a tubular member having one end **12***a*. The one end **12***a* of the

vibration body **12** supports the first surface **11***a* of the light transmissive body **11**. In detail, an end surface of the one end **12***a* of the vibration body **12** supports an outer edge portion of the first surface **11***a* of the light transmissive body **11**. In this preferred embodiment, as illustrated in FIG. **2**B, a surface **12***aa* which supports the first surface **11***a* of the light transmissive body **11** is provided to the one end **12***a* of the vibration body **12**. The outer edge portion of the first surface **11***a* of the light transmissive body **11** is mounted on the surface **12***aa* of the vibration body **12**, and the outer edge portion of the first surface **11***a* of the light transmissive body **11** and the surface **12***aa* of the vibration body **12** is joined to each other, thus the first surface **11***a* of the light transmissive body **11** being supported by the vibration body **12**.

(69) The vibration body **12** receives vibration of the piezoelectric material **14** described later to cause the light transmissive body **11** to vibrate.

(70) As material of the vibration body **12**, metal, such as stainless steel, aluminum, iron, titanium, or duralumin, can be used. In order to reduce loss of vibration transmitted from the piezoelectric material **14** to the light transmissive body **11**, the vibration body **12** is desirably made of material with high rigidity.

(71) In this preferred embodiment, the one end **12***a* of the vibration body **12** and the light transmissive body **11** are fixed to each other by adhesive. In this case, in order to improve adhesion of the adhesive, a surface of the vibration body **12** is preferably applied with oxidation treatment or alumite treatment.

(72) As illustrated in FIG. **2**B, the vibration body **12** has a first threaded portion **17***b* on a side of the one end **12***a*. The first threaded portion **17***b* is threadedly engaged with a second threaded portion **17***a* of the retainer **13**. In this preferred embodiment, the first threaded portion **17***b* is a male thread provided to an outer circumference of the vibration body **12** on the one end **12***a* side.

(73) As illustrated in FIG. **3**, in this preferred embodiment, the vibration body **12** preferably has a cylindrical or substantially cylindrical shape corresponding to the shape of the light transmissive body **11**. In order to efficiently transmit vibration of the piezoelectric material **14** to the light transmissive body **11**, in this preferred embodiment, the vibration body **12** preferably is formed in a shape in which cylinders having diameters different from each other are combined together. In detail, the vibration body **12** is formed such that its outer diameter becomes larger gradually from the one end **12***a* to an other end **12***c*. Since the vibration body **12** preferably has such a shape, vibration of the piezoelectric material **14** can efficiently be transmitted to the light transmissive body **11**.

(74) In this preferred embodiment, a portion of the outer circumference of the vibration body **12** preferably has a hexadecagonal shape, for example. Since the outer circumference of the vibration body **12** is formed in such manner, the vibration body **12** can easily be held by a tool when the first threaded portion **17***b* and the second threaded portion **17***a* are threadedly engaged with each other. Therefore, the first threaded portion **17***b* and the second threaded portion **17***a* can firmly be tightened together. Moreover, numerical control of an amount of tightening torque becomes easier. Note that the shape of the outer circumference of the vibration body **12** is not limited to the hexadecagonal shape, but may be a polygonal shape (for example, an octagonal shape or an icosagonal shape), an oval shape, or a shape having a cut therein for a screwdriver.

(75) Retainer

(76) As illustrated in FIGS. **1** to **3**, the retainer **13** has a side wall **13***f* and the supporting portion **16**. The side wall **13***f* of the retainer **13** has one end **13***a* and an other end **13***b* and surrounds the outer circumference of the light transmissive body **11**. The supporting portion **16** of the retainer **13** extends inwardly with respect to the side wall **13***f* from the one end **13***a*. In this preferred embodiment, as illustrated in FIG. **3**, the retainer **13** preferably has a ring shape. That is, the supporting portion **16** of the retainer **13** is formed to extend inwardly from the one end **13***a* of the side wall **13***f* preferably having a ring shape. As illustrated in FIG. **2**B, the supporting portion **16** is located on the second surface **11***b* side of the light transmissive body **11**, and includes a supporting

surface **16***b* which presses the second surface **11***b* of the light transmissive body **11**. Moreover, the retainer **13** is fixed to the vibration body **12** at a portion in contact with the vibration body **12** on the other end **13***b* side. That is, the retainer **13** and the vibration body **12** are fixed to each other such that the inner side (inner wall portion) of the side wall **13***f* of the retainer **13** contacts the vibration body **12** on the other end **13***b* side of the retainer **13**.
(77) As material of the retainer **13**, metal, such as stainless steel, aluminum, iron, titanium, or duralumin, can be used. A surface of the retainer **13** may be applied with oxidation treatment or alumite treatment similarly to the vibration body **12**.
(78) As illustrated in FIG. **2**B, the supporting portion **16** of the retainer **13** is formed to extend inwardly with respect to the side wall **13***f* from the one end **13***a* of the retainer **13**. The supporting portion **16** has the supporting surface **16***b* which supports the second surface **11***b* of the light transmissive body **11**. The supporting surface **16***b* of the supporting portion **16** of the retainer **13** supports an outer edge portion of the second surface **11***b* of the light transmissive body **11**. In this preferred embodiment, the supporting surface **16***b* of the retainer **13** is adhered to the second surface **11***b* of the light transmissive body **11** by an adhesive **20**.
(79) Since the supporting portion **16** is formed to extend inwardly with respect to the side wall **13***f* from the one end **13***a* of the retainer **13**, the retainer **13** preferably has a circular or substantially circular annular shape when seen from above (in the Y direction).
(80) The supporting portion **16** of the retainer **13** includes a slope surface **16***a* at its tip end. By the supporting portion **16** of the retainer **13** including the slope surface **16***a*, it becomes easier to flow out a water droplet or the like adhering to the light transmissive body **11**. The slope surface **16***a* preferably defines an angle with respect to the second surface **11***b* of the light transmissive body **11** equal to or larger than about 10° and equal to or smaller than about 80°, for example. More preferably, the slope surface **16***a* may define an angle with respect to the second surface **11***b* of the light transmissive body **11** equal to or larger than about 30° and equal to or smaller than about 60°, for example.
(81) In this preferred embodiment, as illustrated in FIG. **2**B, the second threaded portion **17***a* is formed in the retainer **13** on the other end **13***b* side. The second threaded portion **17***a* is threadedly engaged with the first threaded portion **17***b* formed in the vibration body **12**. That is, by the first threaded portion **17***b* and the second threaded portion **17***a* being threadedly engaged to each other, the retainer **13** is fixed to the vibration body **12** on the other end **13***b* side. Being fixed to the vibration body **12** on the other end **13***b* side means that the retainer **13** and the vibration body **12** are fixed to each other at a position closer to the other end **13***b* than to the one end **13***a* of the retainer **13**, or at a position closer to the other end **13***b* than the middle in the height direction (Y direction) of the retainer **13**. In this preferred embodiment, the second threaded portion **17***a* is a female thread formed in the inner wall of the retainer **13** on the other end **13***b* side.
(82) In this preferred embodiment, the retainer **13** preferably has a circular or substantially circular annular shape corresponding to the shape of the light transmissive body **11**. Moreover, as illustrated in FIGS. **1** and **3**, an outer circumference of the retainer **13** preferably has a hexadecagonal shape. Since the outer circumference of the retainer **13** is formed in such manner, the retainer **13** can easily be held by a tool when the first threaded portion **17***b* and the second threaded portion **17***a* are threadedly engaged with each other. Therefore, the first threaded portion **17***b* and the second threaded portion **17***a* can firmly be tightened together. Moreover, numerical control of an amount of tightening torque becomes easier.
(83) Note that the shape of the outer circumference of the retainer **13** is not limited to the hexadecagonal shape, but may be a polygonal shape (for example, an octagonal shape or an icosagonal shape), an oval shape, or a shape having a cut therein for a screwdriver. In the case of having a polygonal shape, the polygonal shape may be an octagon or with more corners so as to maintain vibration performance. In this case, an influence of oscillation due to parasitic vibration which is called “spurious oscillation” can be reduced.

(84) Adhesive

(85) As illustrated in FIG. **2**B, in this preferred embodiment, the adhesive **20** is applied to a joint portion **12***b* between the light transmissive body **11** and the vibration body **12**, a joint portion **13***c* between the light transmissive body **11** and the supporting surface **16***b* of the supporting portion **16** of the retainer **13**, and the joint portion between the first threaded portion **17***a* and the second threaded portion **17***b*. Note that the adhesive **20** is not an essential component, and the application thereof is not always necessary.

(86) As material of the adhesive **20**, hard organic material, such as epoxy resin, can be used. Alternatively, as material of the adhesive **20**, inorganic material, such as cement or a glass frit, may be used. In order to reduce transmission loss of vibration, adhesive with high Young's modulus is preferably used. The adhesive **20** may include a filler having a particle diameter of approximately 10 μm, for example.

(87) By the adhesive **20** being mixed with the filler, when the vibration body **12** and the retainer **13** are fastened to be fixed to each other by the threaded engagement between the first threaded portion **17***a* and the second threaded portion **17***b*, the thickness of the adhesive **20** applied to the joint portions **12***b* and **13***c* can be maintained independent from the tightening torque. By the thickness of the adhesive **20** being maintained, it becomes possible to reduce influence of unevenness of the surface of each component, and improve peel strength.

(88) In this preferred embodiment, the adhesive **20** is filled to the joint portion **12***b* between the light transmissive body **11** and the vibration body **12**, and the joint portion **13***c* between the light transmissive body **11** and the supporting surface **16***b* of the retainer **13**, and the retainer **13** is fastened before the adhesive **20** is hardened. After the retainer **13** is fastened and the first threaded portion **17***b* and the second threaded portion **17***a* are threadedly engaged with each other, the adhesive **20** is hardened. By the joint portions between the light transmissive body **11** and the vibration body **12**, and between the light transmissive body **11** and the retainer **13** being formed as described above, a force to hold the light transmissive body **11** while sandwiching it between the vibration body **12** and the retainer **13** can be maintained. Moreover, in this preferred embodiment, the adhesive **20** is also filled between the first threaded portion **17***a* and the second threaded portion **17***b*, and also plays role as a loosening stopper of the thread.

(89) Piezoelectric Material

(90) The piezoelectric material **14** is fixed to the vibration body **12**. In this preferred embodiment, the piezoelectric material **14** and the vibration body **12** are fixed to each other by adhesive. For example, the piezoelectric material **14** vibrates by being applied with voltage.

(91) As illustrated in FIG. **3**, in this preferred embodiment, the piezoelectric material **14** preferably has a circular or substantially circular annular plate shape. The shape of the piezoelectric material **14** is not limited to a circular or substantially circular annular plate shape and may be any shape that enables vibrating of the vibration body **12**.

(92) As a material forming the piezoelectric material **14**, a suitable piezoelectric ceramic material (for example, barium titanate (BaTiO.sub.3), lead zirconate titanate (PZT: PbTiO.sub.3/PbZrO.sub.3), lead titanate (PbTiO.sub.3), lead metaniobate (PbNb.sub.2O.sub.6), bismuth titanate (Bi.sub.4Ti.sub.3O.sub.12), or (K,Na)NbO.sub.3), or suitable piezoelectric single crystals, such as LiTaO.sub.3 or LiNbO.sub.3, can be used.

(93) As illustrated in FIGS. **2**A and **3**, the piezoelectric material **14** is provided with the conductor **15** to apply voltage to the piezoelectric material **14**. As material of the conductor **15**, metal having high conductivity, such as stainless steel or copper, can be used, for example. Alternatively, the conductor **15** may be wiring formed on flexible printed circuits (FPC). The FPC is a widely used technique, and as a typical example, FPC which is wiring-formed by copper foil on a polyimide substrate is known.

(94) In the case of using the FPC, since it has flexibility, voltage can be applied to the piezoelectric material **14** without impeding vibration.

(95) Comparison with Comparative Example and Reference Example
(96) FIG. **5** is a graph illustrating a time period taken before separation occurs at a joint portion between a light transmissive body and a vibration body in an environment at a temperature of about 85° C. and a humidity of about 85%, for example. In FIG. **5**, a period of time taken before separation in each of a vibration device of Comparative Example 1 which is not provided with a retainer (without retainer), a vibration device of Reference Example 1 provided with a retainer without a supporting portion (retainer pressing obliquely), and the vibration device **10** of Preferred Embodiment 1 (Example 1) (retainer pressing from above) is illustrated. In the vibration device of Reference Example 1, an outer circumferential edge of a light transmissive body is chamfered, and the retainer obliquely supports the outer circumferential edge of the light transmissive body.
(97) As illustrated in FIG. **5**, in the vibration device of Comparative Example 1 (without retainer), in an environment at a temperature of 85° C. and a humidity of 85%, separation between the light transmissive body and the vibration body occurs in approximately ten hours. Moreover, in the vibration device of Reference Example 1 (retainer pressing obliquely), in the same environment, the separation occurs in approximately one hundred hours. On the other hand, in the vibration device **10** of Example 1 (retainer pressing from above), in the same environment, the separation does not occur even after one thousand hours. As described above, by the retainer **13** being provided with the supporting portion **16** having the supporting surface **16***b* to support the second surface **11***b* of the light transmissive body **11**, reliability of the joint portion in the high-temperature and high-humidity environment is largely improved.
(98) Effects
(99) By the vibration device **10** and the imaging device **100** according to Preferred Embodiment 1, the following effects can be achieved.
(100) The vibration device **10** includes the light transmissive body **11**, the vibration body **12**, and the retainer **13**. The light transmissive body **11** has a plate shape including the first surface **11***a* and the second surface **11***b* opposed to the first surface **11***a*. The vibration body **12** is a tubular member including the one end **12***a*, supports the first surface **11***a* of the light transmissive body **11** at the one end **12***a*, and vibrates the light transmissive body **11**. The retainer **13** includes the side wall **13***f* including the one end **13***a* and the other end **13***b* and surrounding the outer circumference of the light transmissive body **11**, and the supporting portion **16** extending inwardly with respect to the side wall **13***f* from the one end **13***a*. The retainer **13** is fixed to the vibration body **12** at the portion in contact with the vibration body **12** on the other end **13***b* side. The supporting portion **16** of the retainer **13** has the supporting surface **16***b* located on the second surface **11***b* side of the light transmissive body **11** and configured to support the second surface **11***b* of the light transmissive body **11**.
(101) In this configuration, the vibration device **10** in which reliability of the joint portion **12***b* between the light transmissive body **11** and the vibration body **12** is improved can be provided. In detail, in the vibration device **10**, the light transmissive body **11** is held by being sandwiched between the retainer **13** and the vibration body **12**, and thus, separation at the joint portion **12***b* between the light transmissive body **11** and the vibration body **12** can be reduced or prevented.
(102) Moreover, by the light transmissive body **11** being held while being sandwiched between the retainer **13** and the vibration body **12**, the light transmissive body **11** can be held firmly. Therefore, vibration from the piezoelectric material **14** is easily transmitted to the light transmissive body **11** through the vibration body **12**, which improves vibration performance of the vibration device **10**.
(103) Moreover, since the light transmissive body **11** is held by being sandwiched between the retainer **13** and the vibration body **12**, a risk that the light transmissive body **11** falls off from the vibration device **10** can largely be reduced.
(104) Moreover, since a force applied from outside to the vibration device **10** is adsorbed by the retainer **13**, a risk of chipping or breaking of the light transmissive body **11** can be reduced. By the outer circumference of the light transmissive body **11** being covered by the retainer **13**, the end

portion of the light transmissive body **11** which is generally weak can be covered. Therefore, it can be suppressed that the end portion of the light transmissive body **11** is locally applied with a large force, and thus, generation of cracks (for example, chipping) of the light transmissive body **11** can be reduced or prevented. As a result, chipping and breaking of the light transmissive body **11** started from the crack can be reduced or prevented.
(105) The one end **12***a* of the vibration body **12** and the first surface **11***a* of the light transmissive body **11** are adhered to each other by the adhesive **20**. The supporting surface **16***b* of the retainer **13** and the second surface **11***b* of the light transmissive body **11** are adhered to each other by the adhesive **20**.
(106) In this configuration, the joint portion between the light transmissive body **11** and the vibration body **12** can be fixed more firmly, and separation can be prevented.
(107) The vibration body **12** includes the first threaded portion **17***b* on the one end **12***a* side. The retainer **13** has the second threaded portion **17***a* formed on the other end **13***b* side and configured to be threadedly engaged with the first threaded portion **17***b*.
(108) In this configuration, the retainer **13** and the vibration body **12** can firmly be fixed to each other.
(109) Moreover, although, in the preferred embodiment described above, the example is described in which the vibration body **12** has a cylindrical or substantially cylindrical shape, the shape is not limited to this. For example, the vibration body **12** may have a polygonal tubular shape or an oval tubular shape.
(110) Moreover, although, in the preferred embodiment described above, the example is described in which the retainer **13** has a circular or substantially circular annular shape, the shape is not limited to this. For example, the retainer **13** may have a polygonal annular shape or an oval annular shape.
(111) Moreover, although, in the preferred embodiment described above, the example is described in which the retainer **13** is made of metal, the material forming the retainer **13** is not limited to this. For example, the retainer **13** may be made of plastic.
(112) Moreover, although, in the preferred embodiment described above, the example is described in which the adhesive **20** is applied to the joint portion **12***b* between the light transmissive body **11** and the vibration body **12**, the joint portion **13***c* between the light transmissive body **11** and the retainer **13**, and the joint portion between the first threaded portion **17***a* and the second threaded portion **17***b*, it is not limited to this. For example, these joint portions may be joined by plating, coating, brazing, or the like. Further, the adhesive **20** only needs to be applied to at least one of between the one end **12***a* of the vibration body **12** and the first surface **11***a* of the light transmissive body **11** or between the supporting surface **16***b* of the retainer **13** and the second surface **11***b* of the light transmissive body **11**.
(113) Moreover, although, in the preferred embodiment described above, the example is described in which the retainer **13** includes the supporting portion **16**, and the one end **12***a* of the vibration body **12** and the supporting portion **16** hold the light transmissive body **11**, it is not limited to this. The light transmissive body **11** only needs to be held by the vibration body **12** and the retainer **13** while being sandwiched therebetween.
(114) Modification
(115) FIG. **6** is a sectional view of a vibration device **10**A according to Modification 1 of Preferred Embodiment 1. As illustrated in FIG. **6**, a light transmissive body **111** may have a dome shape. The dome shape is a shape in which a plate-shaped member is curved hemispherically, for example. In this case, the light transmissive body **111** has, at its end portion, a first surface **111***a* and a second surface **111***b* opposed to the first surface **111***a*.
(116) Alternatively, the light transmissive body may have a plate shape in a polygonal shape (for example, a quadrangular shape, a hexagonal shape, or an octagonal shape), or an oval shape.
Preferred Embodiment 2

(117) A vibration device according to Preferred Embodiment 2 of the present invention is described. Note that, in Preferred Embodiment 2, points different from Preferred Embodiment 1 are mainly described. In Preferred Embodiment 2, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 1. Also, in Preferred Embodiment 2, descriptions overlapping those in Preferred Embodiment 1 are omitted.
(118) FIG. **7** is a partial sectional view illustrating a vibration device **110** according to Preferred Embodiment 2.
(119) In Preferred Embodiment 2, the shape of a supporting portion **116** of a retainer **113** is different from the shape in Preferred Embodiment 1.
(120) As illustrated in FIG. **7**, a length L**1** of the supporting portion **116** of the retainer **113** is shorter than the length of the supporting portion **16** of the retainer **13** of the vibration device **10** according to Preferred Embodiment 1. Here, as illustrated in FIG. **7**, the length L**1** of the supporting portion **116** of the retainer **113** indicates a length in a direction from one end **113***a* of the retainer **113** to an inner side with respect to a side wall **113***f*.
(121) As described above, by the length L**1** of the supporting portion **116** of the retainer **113** being made shorter, a supporting surface **116***b* supports the light transmissive body **11** at a portion closer to the outer edge. By the light transmissive body **11** being supported at the portion closer to the outer edge, a field of view during imaging by the imaging part disposed inside can be widened.
(122) FIG. **8** is a graph illustrating relationship between a position where the supporting surface **116***b* of the retainer **113** supports the light transmissive body **11** and compressive stress. A torque notch position indicated by a horizontal axis illustrates a position in a direction toward an outer side portion of the light transmissive body **11** by setting a position of a broken line P**0** in FIG. **7** as zero. The position of the broken line P**0** is the most inner side position of the joint portion **12***b* between the first surface **11***a* of the light transmissive body **11** and the one end **12***a* of the vibration body **12**. In this preferred embodiment, a position of a broken line P**3** is about 3 mm, for example.
(123) As illustrated in the graph in FIG. **8**, as the position where the supporting surface **116***b* of the retainer **113** supports the second surface **11***b* of the light transmissive body **11** is a more inner side portion of the light transmissive body **11**, compressive stress becomes larger. The compressive stress becoming larger indicates that separation at the joint portion **12***b* between the light transmissive body **11** and the one end **12***a* of the vibration body **12** becomes less likely to occur.
(124) On the other hand, when the supporting surface **116***b* is provided to the outer side portion of the light transmissive body **11**, the field of view during imaging by the imaging part can be widened. Therefore, the supporting surface **116***b* of the retainer **113** may be located outside the position of the broken line P**0**.
(125) FIG. **9** is a schematic sectional view illustrating an inner diameter of the supporting portion **116** of the retainer **113**, an inner diameter of the vibration body **12**, and an outer diameter of the light transmissive body **11** in the vibration device **110** in FIG. **7**. In order to achieve both of improving the compressive stress and widening the field of view, an inner diameter C of the supporting portion **116** of the retainer **113** may be equal to or larger than an inner diameter A of the vibration body **12** and smaller than one-half of the sum of the inner diameter A of the vibration body **12** and an outer diameter B of the light transmissive body **11**. That is, the length L**1** of the supporting portion **116** of the retainer **113** may be determined such that the relationship "$A \leq C < (A+B)/2$" is satisfied.
(126) Effects
(127) By the vibration device **110** according to Preferred Embodiment 2, following effects can be achieved.
(128) The light transmissive body **11** has a circular or substantially circular plate shape. The vibration body **12** and the retainer **113** have a cylindrical or substantially cylindrical shape. The inner diameter C of the supporting portion **116** of the retainer **113** is equal to or larger than the inner diameter A of the vibration body **12** and smaller than one-half of the sum of the inner

diameter A of the vibration body **12** and the outer diameter B of the light transmissive body **11**.
(129) In this configuration, the field of view of the imaging part can be widened while reliability of the joint portion is improved. The reliability of the joint portion means unlikeliness of separation and high strength at the joint portion **12***b* between the light transmissive body **11** and the vibration body **12**. In the preferred embodiment described above, since the light transmissive body **11** can firmly be held by the retainer **113** and the vibration body **12**, reliability of the joint portion can be improved, that is, the occurrence of separation can be reduced, and the strength of the joint portion can be improved. Further, compared to the vibration device **10** in Preferred Embodiment 1, the supporting surface **116***b* of the retainer **113** is provided to the more outer side portion of the light transmissive body **11**. Therefore, the field of view during imaging can be widened.
Preferred Embodiment 3
(130) A vibration device according to Preferred Embodiment 3 of the present invention is described. Note that, in Preferred Embodiment 3, points different from Preferred Embodiment 2 are mainly described. In Preferred Embodiment 2, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 2. Also, in Preferred Embodiment 3, descriptions overlapping those in Preferred Embodiment 2 are omitted.
(131) FIG. **10** is a sectional view of a vibration device **110**A according to Preferred Embodiment 3.
(132) As illustrated in FIG. **10**, Preferred Embodiment 3 is different from Preferred Embodiment 2 in that an inner diameter F of the supporting portion **116** of the retainer **113** is smaller than an inner diameter D of the vibration body **12**. In this case, the inner diameter F of the supporting portion **116** of the retainer **113** may be smaller than about one-half of the sum of the inner diameter D of the vibration body **12** and an outer diameter E of the light transmissive body **11**.
(133) In this case, since a contact area between the retainer **113** and the light transmissive body **11** increases, the light transmissive body **11** can be fixed more firmly.
Preferred Embodiment 4
(134) A vibration device according to Preferred Embodiment 4 of the present invention is described. Note that, in Preferred Embodiment 4, points different from Preferred Embodiment 1 are mainly described. In Preferred Embodiment 4, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 1. Also, in Preferred Embodiment 4, descriptions overlapping those in Preferred Embodiment 1 are omitted.
(135) FIG. **11** is a partial sectional view of a vibration device **210** according to Preferred Embodiment 4.
(136) Preferred Embodiment 4 is different from Preferred Embodiment 1 in that a supporting portion **216** of a retainer **213** includes a first supporting portion **216***c* and a second supporting portion **216***d*.
(137) The first supporting portion **216***c* extends inwardly with respect to a side wall **213***f* from one end **213***a* of the retainer **213**. A space sp is formed between the first supporting portion **216***c* and the second surface **11***b* of the light transmissive body **11**.
(138) The second supporting portion **216***d* projects toward the second surface **11***b* of the light transmissive body **11** from a tip end of the first supporting portion **216***c*, and has a supporting surface **216***b*.
(139) The space sp between the first supporting portion **216***c* and the second surface **11***b* of the light transmissive body **11** may be formed by, for example, a recess being provided to the supporting surface **16***b* of the supporting portion **16** of the retainer **13** described in Preferred Embodiment 1 (FIG. **2**B). Alternatively, the space sp may be formed by the first supporting portion **216***c* being formed such that a thickness h**1** of the first supporting portion **216***c* of the retainer **213** is smaller than a thickness h**2** of the second supporting portion **216***d*. Note that, in this preferred embodiment, the thickness h**1** of the first supporting portion **216***c* is about 0.7 mm, and the thickness h**2** of the second supporting portion **216***d* is about 0.8 mm, for example.
(140) A width w**1** of the second supporting portion **216***d* of the retainer **213** in a direction in which

the supporting portion **216** extends (that is, the width w**1** of the supporting surface **216***b*) may be about 0.3 times or larger and about 3 times or smaller the thickness h**1** of the first supporting portion **216***c*, for example. That is, the width w**1** of the supporting surface **216***b* of the retainer **213** may be about 0.3 times or larger and about 3 times or smaller the thickness h**1** of the first supporting portion **216***c*, for example. In this preferred embodiment, the thickness h**1** of the first supporting portion **216***c* is about 0.7 mm, and the width w**1** of the second supporting portion **216***d* is about 1 mm, for example.
(141) By the supporting portion **216** of the retainer **213** being formed in this manner, the supporting surface **216***b* provided to the second supporting portion **216***d* can support the light transmissive body **11** at the more inner side portion.
(142) Effects
(143) By the vibration device **210** according to Preferred Embodiment 4, following effects can be achieved.
(144) The supporting portion **216** of the retainer **213** includes the first supporting portion **216***c* and the second supporting portion **216***d*. The first supporting portion **216***c* extends inwardly with respect to the side wall **213***f* from the one end **213***a* of the retainer **213**. The second supporting portion **216***d* projects toward the second surface **11***b* of the light transmissive body **11** from the tip end of the first supporting portion **216***c*, and has the supporting surface **216***b*. The space is formed between the first supporting portion **216***c* and the second surface **11***b* of the light transmissive body **11**. The width w**1** of the second supporting portion **216***d* of the retainer **213** in the extending direction of the supporting portion **216** is about 0.3 times or larger and about 3 times or smaller the thickness h**1** of the first supporting portion **216***c*, for example.
(145) In this configuration, the vibration device **210** with higher reliability can be provided. The separation between the light transmissive body **11** and the vibration body **12** occurs from the inner side portion of the light transmissive body **11** where amplitude is large. However, by the second supporting portion **216***d* of the supporting portion **216** supporting the second surface **11***b* of the light transmissive body **11**, the more inner side portion of the light transmissive body **11** can be fixed.
(146) In the vibration device **10** in Preferred Embodiment 1, the outer edge portion of the second surface lib of the light transmissive body **11** is supported by the substantially entire supporting portion **16** of the retainer **13**. On the other hand, in this preferred embodiment, the outer edge portion of the second surface **11***b* of the light transmissive body **11** is supported by the second supporting portion **216***d* of the retainer **213**. Therefore, a contact area between the supporting surface **216***b* provided to the second supporting portion **216***d* of the retainer **213** and the second surface **11***b* of the light transmissive body **11** in Preferred Embodiment 4 becomes smaller than the contact area between the supporting surface **16***b* of the retainer **13** and the second surface **11***b* of the light transmissive body **11** in Preferred Embodiment 1. The smaller the contact area is, the larger the force applied to the contact portion becomes. In Preferred Embodiment 4, since the second supporting portion **216***d* is formed at the tip end of the first supporting portion **216***c*, the supporting surface **216***b* supports the light transmissive body **11** at the more inner side portion compared to Preferred Embodiment 1. Therefore, in the case of the supporting surface **216***b* of the vibration device **210** in this preferred embodiment, the force pressing the light transmissive body **11** is applied to the more inner side portion of the light transmissive body **11** compared to Preferred Embodiment 1. By the more inner side portion being fixed, separation between the light transmissive body **11** and the vibration body **12** is reduced, and reliability of the joint portion can be improved.
Preferred Embodiment 5
(147) A vibration device according to Preferred Embodiment 5 of the present invention is described. Note that, in Preferred Embodiment 5, points different from Preferred Embodiment 1 are mainly described. In Preferred Embodiment 5, the same reference characters are given to the same

or equivalent components as or to those of Preferred Embodiment 1. Also, in Preferred Embodiment 5, descriptions overlapping those in Preferred Embodiment 1 are omitted.
(148) FIG. **12** is a perspective view illustrating a vibration device **310** according to Preferred Embodiment 5. FIG. **13** is a partial sectional view illustrating the vibration device **310** in FIG. **12**. FIG. **14** is an exploded perspective view illustrating the vibration device **310** in FIG. **12**.
(149) Preferred Embodiment 5 is different from Preferred Embodiment 1 in that a retainer **313** and a vibration body **312** are fixed to each other by a plurality of screws **18**.
(150) As illustrated in FIGS. **12** to **14**, in the vibration device **310**, the retainer **313** and the vibration body **312** are fixed to each other by four screws **18** instead of by the first threaded portion and the second threaded portion. Four through-holes **313***d* are provided from an end surface of the retainer **313** on a side of one end **313***a* to the other end. Four threaded-holes **312***d* are also provided to an end surface of the vibration body **312** on a side of the one end **312***a*.
(151) In this preferred embodiment, the four screws **18** are disposed at an equal interval. By the four screws being disposed at an equal interval, the retainer **313** and the vibration body **312** can stably be fixed to each other. Therefore, the light transmissive body **11** can firmly be fixed while being sandwiched between the supporting portion **316** of the retainer **313** and the vibration body **12**.
(152) Effects
(153) By the vibration device **310** according to Preferred Embodiment 5, following effects can be achieved.
(154) The retainer **313** and the vibration body **312** are fixed to each other by the plurality of screws **18**.
(155) In this configuration, the retainer **313** and the vibration body **312** can be fixed to each other at lower cost than the first threaded portion and the second threaded portion respectively being formed in the retainer **313** and the vibration body **312**.
(156) Note that, although, in Preferred Embodiment 5, the example is described in which the retainer **313** and the vibration body **312** are fixed to each other by the four screws **18**, the number of screws is not limited to four as long as the number of screws is two or more. In this case, the screws may be disposed at an equal interval.
(157) Moreover, the vibration body **312** may have four through-holes instead of the four threaded-holes **312***d*. In this case, the retainer **313** and the vibration body **312** may be fixed to each other by a bolt being inserted into the through-holes of the retainer **313** and the vibration body **312**, and fastened by a nut.
Preferred Embodiment 6
(158) A vibration device according to Preferred Embodiment 6 of the present invention is described. Note that, in Preferred Embodiment 6, points different from Preferred Embodiment 1 are mainly described. In Preferred Embodiment 6, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 1. Also, in Preferred Embodiment 6, descriptions overlapping those in Preferred Embodiment 1 are omitted.
(159) FIG. **15** is an exploded perspective view illustrating a vibration device **410** according to Preferred Embodiment 6. FIG. **16** is a partial sectional view of the vibration device **410** in FIG. **15**.
(160) Preferred Embodiment 6 is different from Preferred Embodiment 1 in that a supporting portion **416** of a retainer **413** is a plate spring.
(161) As illustrated in FIGS. **15** and **16**, the supporting portion **416** of the retainer **413** is a plate spring. In this preferred embodiment, as illustrated in FIG. **15**, the supporting portion **416** of the retainer **413** includes an annular plate spring.
(162) As illustrated in FIG. **16**, the retainer **413** is formed by a member in a plate shape, and the supporting portion **416** of the retainer **413** is formed to be curved inwardly at one end **413***a* of the retainer **413**. In detail, the supporting portion **416** includes a spring portion **416***c* curved inwardly at the one end **413***a* of the retainer **413**, and a pressing surface **416***d* supporting the second surface

11*b* of the light transmissive body **11**. Being curved inwardly means to be curved toward the second surface **11***b* of the light transmissive body **11** from the one end **413***a* of the retainer **413**.
(163) The supporting portion **416** presses the second surface **11***b* of the light transmissive body **11** by restoring force of the plate spring. Further, the supporting portion **416** and the second surface **11***b* of the light transmissive body **11** may be adhered to each other by adhesive.
(164) A joint portion **417** between the retainer **413** and a vibration body **412** is fixed by welding, adhesive, or the like.
(165) Effects
(166) By the vibration device **410** according to Preferred Embodiment 6, following effects can be achieved.
(167) The supporting portion **416** of the retainer **413** is one or a plurality of plate springs.
(168) In this configuration, manufacturing cost of the retainer **413** can be reduced.
(169) Note that, although, in Preferred Embodiment 6, the example is described in which the supporting portion **416** is an annular plate spring, the shape of the supporting portion **416** is not limited to this. For example, the supporting portion **416** may include a plurality of plate springs. In this case, the plate springs may be disposed at an equal interval.
Preferred Embodiment 7
(170) A vibration device according to Preferred Embodiment 7 of the present invention is described. Note that, in Preferred Embodiment 7, points different from Preferred Embodiment 4 are mainly described. In Preferred Embodiment 7, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 4. Also, in Preferred Embodiment 7, descriptions overlapping those in Preferred Embodiment 4 are omitted.
(171) FIG. **17** is an exploded perspective view illustrating a vibration device **510** according to Preferred Embodiment 7. FIG. **18** is a partial sectional view of the vibration device **510** in FIG. **17**.
(172) Preferred Embodiment 7 is different from Preferred Embodiment 1 in that a groove **516***e* is formed in a supporting portion **516** of a retainer **513**, and a sealing member **19** is provided to the groove **516***e*.
(173) As illustrated in FIG. **17**, the sealing member **19** is disposed between the retainer **513** and the second surface **11***b* of the light transmissive body **11**. By the sealing member **19** being provided, waterproofing can be achieved between the retainer **513** and the second surface **11***b* of the light transmissive body **11**, and entering of foreign matters, such as a water droplet or the like, into the vibration device **510** can be prevented.
(174) As illustrated in FIG. **18**, the groove **516***e* is formed in a surface **516***b* of the supporting portion **516** of the retainer **513**, the surface **516***b* being opposed to the second surface **11***b* of the light transmissive body **11**. The groove **516***e* is formed in an annular shape and is continuous along an inner circumference of the retainer **513**. Although, in this preferred embodiment, the groove **516***e* is formed in a circular annular shape, it may be formed in a polygonal shape or an oval shape, for example.
(175) The sealing member **19** is an elastic member such as an O ring, for example. The sealing member **19** may be any member as long as it can achieve waterproofing between the retainer **513** and the second surface **11***b* of the light transmissive body **11**.
(176) Effects
(177) By the vibration device **510** according to Preferred Embodiment 7, following effects can be achieved.
(178) The groove **516***e* is formed to be continuous along the inner circumference of the retainer **513** on the surface **516***b* of the supporting portion **516** of the retainer **513**, the surface **516***b* being opposed to the second surface **11***b* of the light transmissive body **11**. The sealing member **19** is provided to the groove **516***e*.
(179) As the sealing member **19**, an O ring made of rubber can be used, for example. O rings are generally higher in waterproof performance than adhesive, and particularly, also in a high-

temperature and high-humidity environment at a temperature of about 85° C. and a humidity of about 85%, for example, entering of water vapor can be suppressed. Therefore, by the sealing member **19** being provided between the supporting portion **516** of the retainer **513** and the second surface **11***b* of the light transmissive body **11**, moisture-resistance reliability of the vibration device **510** can be improved.

Preferred Embodiment 8

(180) A vibration device according to Preferred Embodiment 8 of the present invention is described. Note that, in Preferred Embodiment 8, points different from Preferred Embodiment 1 are mainly described. In Preferred Embodiment 8, the same reference characters are given to the same or equivalent components as or to those of Preferred Embodiment 1. Also, in Preferred Embodiment 8, descriptions overlapping those in Preferred Embodiment 1 are omitted.

(181) FIG. **19** is a partial sectional view illustrating a vibration device **610** according to Preferred Embodiment 8.

(182) Preferred Embodiment 8 is different from Preferred Embodiment 1 in that a light transmissive body **611** includes a body portion **611***c* and a frame portion **611***d*.

(183) As illustrated in FIG. **19**, the light transmissive body **611** includes the body portion **611***c*, and the frame portion **611***d* provided to an outer edge of the body portion **611***c* and having a thickness h**4** smaller than a thickness h**3** of the body portion **611***c*.

(184) A supporting portion **616** of a retainer **613** supports the frame portion **611***d*. That is, the supporting portion **616** of the retainer **613** is provided to the frame portion **611***d* having the smaller thickness. The frame portion **611***d* of the light transmissive body **611** has a surface formed at a position lower than a second surface **611***b* of the body portion **611***c* in the Y direction. The body portion **611***c* and the frame portion **611***d* are formed integrally. By the supporting portion **616** of the retainer **613** being provided to the frame portion **611***d*, the second surface **611***b* of the light transmissive body **611** and an upper surface of the retainer **613** become flat.

(185) The thickness h**4** of the frame portion **611***d* may be one-tenth or larger and nine-tenth or smaller of the thickness h**3** of the body portion **611***c*. By the thickness h**4** of the frame portion **611***d* being made in this range, strength of the frame portion **611***d* is maintained, and damage of the light transmissive body **611** can be prevented. Further, the field of view during imaging by the imaging part can be widened.

(186) Effects

(187) By the vibration device **610** according to Preferred Embodiment 8, following effects can be achieved.

(188) The light transmissive body **611** includes the body portion **611***c* and the frame portion **611***d*. The frame portion **611***d* is provided to the outer edge of the body portion **611***c*, and has the thickness h**4** smaller than the thickness h**3** of the body portion **611***c*.

(189) In this configuration, when the retainer **613** and the vibration body **12** are fixed to each other, the second surface **611***b* of the body portion **611***c* of the light transmissive body **11** and the upper surface of the supporting portion **616** of the retainer **613** become flat. Therefore, foreign matters, such as a water droplet or the like, adhering to the light transmissive body **11** easily flow out. Further, the field of view during imaging by the imaging part can be widened. Note that it is not always necessary that the second surface **611***b* of the light transmissive body **11** and the upper surface of the retainer **613** are flat. The effect to avoid impedance of the field of view can sufficiently be achieved as long as the upper surface of the retainer **613** does not excessively project with respect to the second surface lib of the light transmissive body **611**.

Examples

(190) Values of compressive stress applied to the vibration device are calculated through simulation using the vibration device **10** of Preferred Embodiment 1 as Example 2 and the vibration device **210** of Preferred Embodiment 4 as Example 3. Moreover, as Reference Example 2, compressive stress is measured similarly to Example 2 and Example 3, using a vibration device provided with a

retainer without a supporting portion.
(191) In the simulation, piezoelectric analysis and stress analysis are performed using Femtet® which is analysis software manufactured by Murata Software Co., Ltd.
(192) FIG. **20**A is a diagram illustrating distribution of compressive stress of the vibration device **10** of Example 2. FIG. **20**B is a diagram illustrating distribution of compressive stress of the vibration device **210** of Example 3. FIG. **20**C is a diagram illustrating distribution of compressive stress of a vibration device of Reference Example 2. In FIGS. **20**A to **20**C, a light-color part indicates a smaller compressive stress, and a dark-color part indicates a larger compressive stress.
(193) Vibration Device of Example 2
(194) Configuration of the vibration device **10** of Example 2 is described in Preferred Embodiment 1, and thus description thereof is omitted.
(195) Vibration Device of Example 3
(196) Configuration of the vibration device **210** of Example 3 is described in Preferred Embodiment 4, and thus description thereof is omitted.
(197) Vibration Device of Reference Example 2
(198) The vibration device of Reference Example 2 includes a light transmissive body having a slope surface at an outer edge portion thereof, and a retainer which supports the slope surface of the light transmissive body. Other configurations are the same as the vibration device **10** of Example 2.
(199) Comparison Result of Compressive Stress
(200) Based on the simulation result of the vibration device **10** of Example 2 illustrated in FIG. **20**A, the compressive stress concentrates on the first surface **11***a* of the light transmissive body **11**. In this case, the compressive stress at the joint portion between the light transmissive body **11** and the vibration body **12** is 2.89 MPa.
(201) Based on the simulation result of the vibration device **210** of Example 3 illustrated in FIG. **20**B, similarly to Example 2, the compressive stress concentrates on the first surface **11***a* of the light transmissive body **11**. In this case, the compressive stress at the joint portion between the light transmissive body **11** and the vibration body **12** is 3.18 MPa. In Example 3, the second supporting portion **216***d* of the retainer **213** supports the more inner side portion of the light transmissive body **11** compared to Example 2. Therefore, the light transmissive body **11** is firmly fixed at the more inner side portion compared to Example 2. Thus, in the vibration device **210** of Example 3, the compressive stress at the joint portion is larger than in Example 2.
(202) Based on the simulation result of the vibration device of Reference Example 2 illustrated in FIG. **20**C, the compressive stress at the first surface of the light transmissive body is smaller than in Example 2 and Example 3. Therefore, in the vibration device of Reference Example 2, separation is likely to occur at the joint portion between the light transmissive body and the vibration body. In this case, the compressive stress at the joint portion between the light transmissive body and the vibration body is 0.86 MPa.
(203) Based on the results described above, it is clear that the compressive stress at the joint portion between the light transmissive body and the vibration body becomes higher when the retainer supports the upper surface (second surface) of the light transmissive body than when the retainer supports the side surface (slope surface) of the light transmissive body. As the compressive stress increases, occurrence of the separation at the joint portion between the light transmissive body and the vibration body can be suppressed, which improves reliability of the vibration device.
(204) Moreover, when comparing the results of Example 2 and Example 3, the compressive stress at the joint portion between the light transmissive body and the vibration body is higher in the vibration device **210** of Example 3. The separation at the joint portion between the light transmissive body and the vibration body is likely to occur from the inner side portion of the light transmissive body where amplitude is large. Therefore, by the second supporting portion being provided like in Example 3, the more inner side portion of the light transmissive body can be supported. Thus, reliability of the vibration device can be improved more in the configuration of

Example 3.
(205) Note that the present disclosure includes combination of arbitrary preferred embodiments and/or examples among the various embodiments and/or examples described above, and can achieve the effect of each preferred embodiment and/or example.
(206) The vibration devices and the imaging devices according to various preferred embodiments of the present invention and modifications and combinations thereof are applicable to car-mounted cameras, surveillance cameras, or optical sensors such as LiDAR, which are used outdoors.
(207) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A vibration device comprising: a tubular vibration body; a retainer fixed to the vibration body; and a light transmissive body to be vibrated by the vibration body and held between the vibration body and the retainer; wherein the light transmissive body includes a first surface and a second surface opposed to the first surface; the vibration body includes one end and supports the first surface of the light transmissive body at the one end; the vibration body includes an outer surface opposed to the one end; and the outer surface of the vibration body is not in contact with the retainer.
2. The vibration device according to claim 1, wherein the retainer includes: a side wall including a first end and a second end and surrounding an outer circumference of the light transmissive body; and a supporting portion extending inwardly with respect to the side wall from the first end of the side wall; the retainer is fixed to the vibration body at a portion in contact with the vibration body on a side of the second end; and the supporting portion includes a supporting surface on a side of the second surface of the light transmissive body to support the second surface of the light transmissive body.
3. The vibration device according to claim 2, wherein at least one of adhesion between the one end of the vibration body and the first surface of the light transmissive body or adhesion between the supporting surface of the retainer and the second surface of the light transmissive body is provided by adhesive.
4. The vibration device according to claim 2, wherein the supporting portion of the retainer includes: a first supporting portion extending inwardly with respect to the side wall from the one end of the retainer; and a second supporting portion projecting toward the second surface of the light transmissive body from a tip end of the first supporting portion and including the supporting surface; and a space is provided between the first supporting portion and the second surface of the light transmissive body.
5. The vibration device according to claim 4, wherein a width of the second supporting portion of the retainer in a direction in which the supporting portion extends is about 0.3 times or larger and about 3 times or smaller a thickness of the first supporting portion.
6. The vibration device according to claim 2, wherein the light transmissive body has a circular or substantially circular plate shape; the vibration body and the retainer have a cylindrical or substantially cylindrical shape; and an inner diameter of the supporting portion of the retainer is equal to or larger than an inner diameter of the vibration body and smaller than about one-half of a sum of the inner diameter of the vibration body and an outer diameter of the light transmissive body.
7. The vibration device according to claim 2, wherein the light transmissive body has a circular or substantially circular plate shape; the vibration body and the retainer have a cylindrical or substantially cylindrical shape; and an inner diameter of the supporting portion of the retainer is

smaller than an inner diameter of the vibration body and smaller than about one-half of a sum of the inner diameter of the vibration body and an outer diameter of the light transmissive body.

8. The vibration device according to claim 2, wherein the retainer includes a second threaded portion on the side of the second end; and the vibration body includes a first threaded portion on a side of the one end of the vibration body to be threadedly engaged with the second threaded portion.

9. The vibration device according to claim 2, wherein the retainer and the vibration body are fixed to each other by a plurality of screws.

10. The vibration device according to claim 2, wherein the supporting portion of the retainer includes one or a plurality of plate springs.

11. The vibration device according to claim 2, wherein a groove extends continuously along an inner circumference of the retainer in a surface of the supporting portion of the retainer opposed to the second surface of the light transmissive body; and a seal is provided to the groove.

12. The vibration device according to claim 2, wherein the light transmissive body includes: a body portion; and a frame portion on an outer edge of the body portion and having a thickness smaller than a thickness of the body portion; the supporting portion of the retainer supports the frame portion; and a thickness of the frame portion is about one-tenth or larger and about nine-tenth or smaller of a thickness of the body portion.

13. An imaging device comprising: the vibration device according to claim 1; and an image pickup device inside the vibration device.

14. The imaging device according to claim 13, wherein the retainer includes: a side wall including a first end and a second end and surrounding an outer circumference of the light transmissive body; and a supporting portion extending inwardly with respect to the side wall from the first end of the side wall; the retainer is fixed to the vibration body at a portion in contact with the vibration body on a side of the second end; and the supporting portion includes a supporting surface on a side of the second surface of the light transmissive body to support the second surface of the light transmissive body.

15. The imaging device according to claim 14, wherein at least one of adhesion between the one end of the vibration body and the first surface of the light transmissive body or adhesion between the supporting surface of the retainer and the second surface of the light transmissive body is provided by adhesive.

16. The imaging device according to claim 14, wherein the supporting portion of the retainer includes: a first supporting portion extending inwardly with respect to the side wall from the one end of the retainer; and a second supporting portion projecting toward the second surface of the light transmissive body from a tip end of the first supporting portion and including the supporting surface; and a space is provided between the first supporting portion and the second surface of the light transmissive body.

17. The imaging device according to claim 16, wherein a width of the second supporting portion of the retainer in a direction in which the supporting portion extends is about 0.3 times or larger and about 3 times or smaller a thickness of the first supporting portion.

18. The imaging device according to claim 14, wherein the light transmissive body has a circular or substantially circular plate shape; the vibration body and the retainer have a cylindrical or substantially cylindrical shape; and an inner diameter of the supporting portion of the retainer is equal to or larger than an inner diameter of the vibration body and smaller than about one-half of a sum of the inner diameter of the vibration body and an outer diameter of the light transmissive body.

19. The imaging device according to claim 14, wherein the light transmissive body has a circular or substantially circular plate shape; the vibration body and the retainer have a cylindrical or substantially cylindrical shape; and an inner diameter of the supporting portion of the retainer is smaller than an inner diameter of the vibration body and smaller than about one-half of a sum of

the inner diameter of the vibration body and an outer diameter of the light transmissive body.

20. The imaging device according to claim 14, wherein the retainer includes a second threaded portion on the side of the second end; and the vibration body includes a first threaded portion on a side of the one end of the vibration body to be threadedly engaged with the second threaded portion.